SNPs and STRs in human geome

Single Nucleotide Polymorphisms and Short Tandem Repeats

UZH-BIO392
Feifei Xia
feifei.xia@uzh.ch

Types of Human Genetic Variants

Differences in DNA among individuals drive many types of phenotypic differences.
There are many types of genomic differences between individuals

- Structural variations (SVs)
- Copy number variations (CNVs)
- **Single nucleotide polymorphisms (SNPs)**
- Short tandem repeats (STRs)

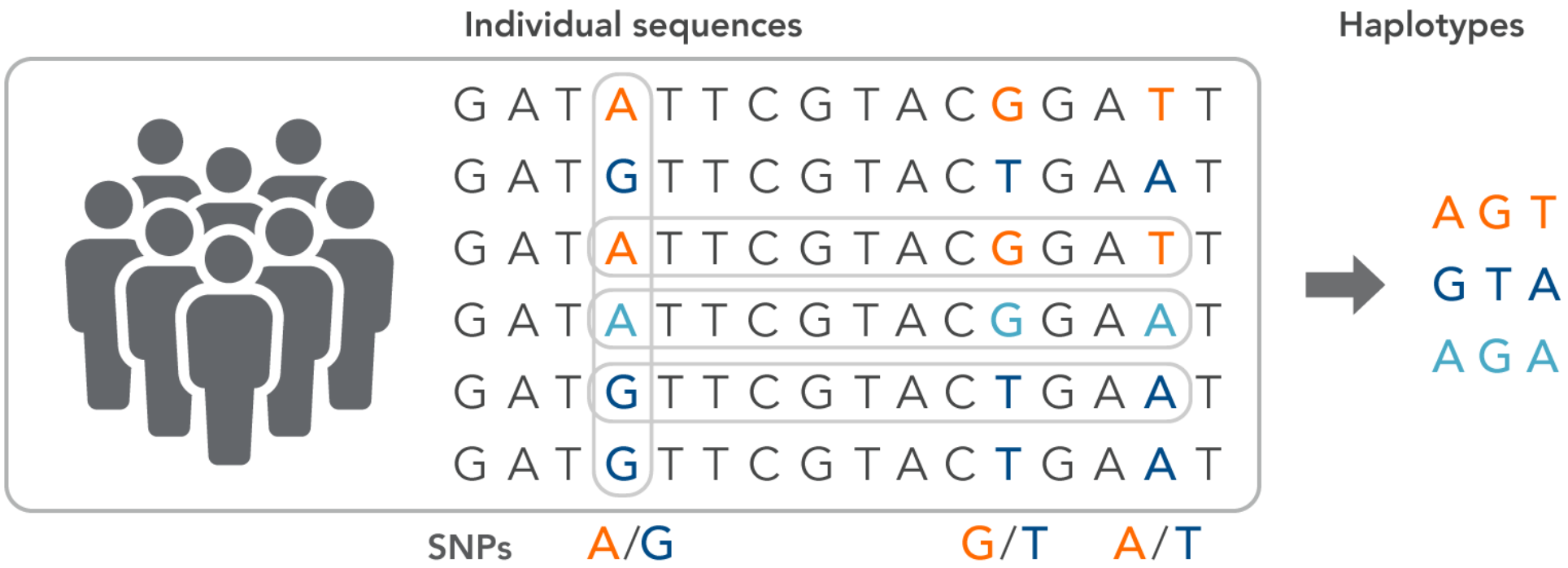
What are SNPs?

- SNPs are the most common type of genetic variation in the human genome. They represent a single base change (A, T, C, or G) at a specific position in the DNA sequence
- We distinguish between SNPs which are relatively common (occurring in at least 1% of the population) which are called single nucleotide polymorphisms or SNPs, and rarer variants called SNVs.

```
Reference CCGTTAGAGTTACAATTCGA
Read 2          TTAGAGTAACAA
Read 3      CCGTTAGAGTTA
Read 4                  TTACAATTCGA
Read 5              GAGTAACAA
Read 6          TTAGAGTAACAAT
```

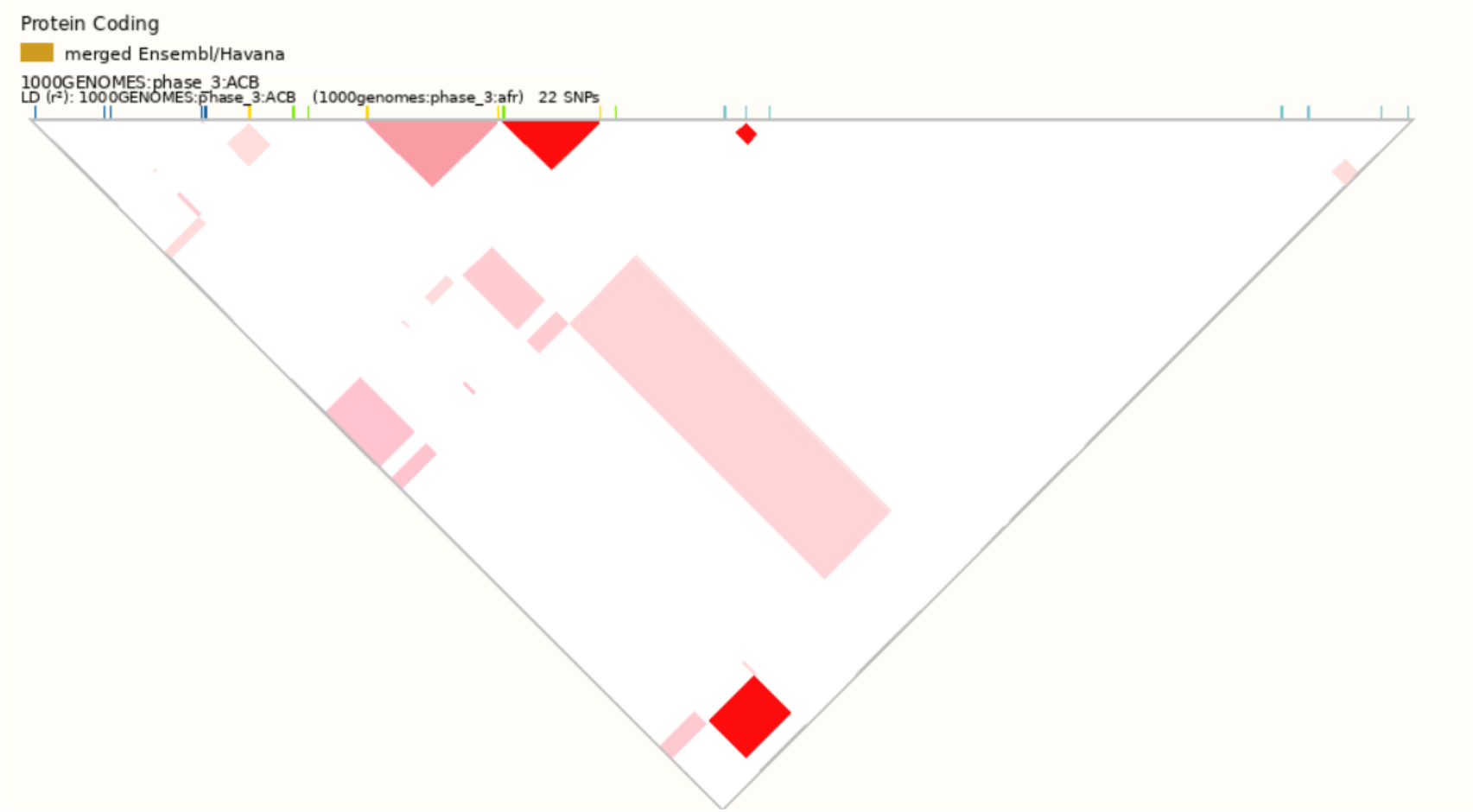
SNP Haplotype

- A SNP haplotype is a set of closely linked SNPs on the same chromosome that are inherited together as a block from a single parent.
- It represents a specific combination of alleles at multiple SNP positions.



Linkage Disequilibrium

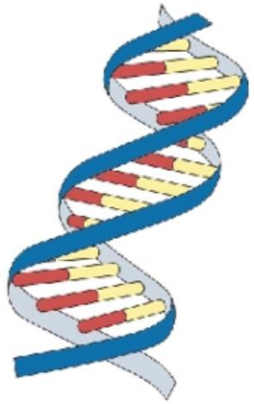
LD refers to the non-random association of alleles at different loci in a population.

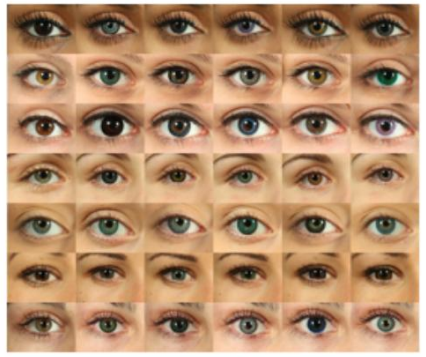


https://www.ensembl.org/Help/View?id=197

Genotypes to phenotypes

- Most SNPs have no effect on health or development. However, some of them have proven to be very important in the study of human health.
- Genetic variation between individuals leads to differences in an individual's **phenotype**, trait or risk of developing a disease. An individual's phenotype is influenced both by their genotype and their environment



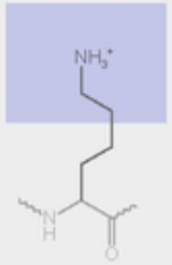
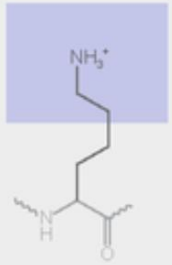
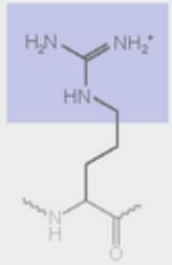
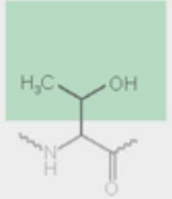


Genotypes are the genetic make-up of an individual.

Phenotypes are the physical traits and characteristics of an individual and are influenced by their genotype and the environment.

What effect do SNPs in coding regions have?

When SNPs occur within a gene or in a regulatory region near a gene, they may play a more direct role in disease by affecting the gene's function.

	No mutation	Point mutations			
		Silent	Nonsense	Missense	
				conservative	non-conservative
DNA level	TTC	TTT	ATC	TCC	TGC
mRNA level	AAG	AAA	UAG	AGG	ACG
protein level	Lys	Lys	STOP	Arg	Thr
					

SNP Genotyping Methods

SNP genotyping are techniques that analyze genomic sequence variations. These single-base substitutions are typically detected using real-time PCR, microarrays, or next-generation sequencing (NGS) techniques.

Real-Time PCR

- High sensitivity
- Familiar workflow
- Capital equipment available in most labs

Microarrays

- Trusted data quality and proven track record
- Scalable and cost-effective for detecting known SNPs in large population studies
- Can focus on target-rich areas
- Robust data analysis pipeline and uncomplicated variant interpretation

NGS

- Unbiased variant discovery using only limited knowledge of the presence or nature of the variants
- High-throughput capabilities and scalability
- Small sample size input
- Emerging user-friendly tools to simplify data analysis

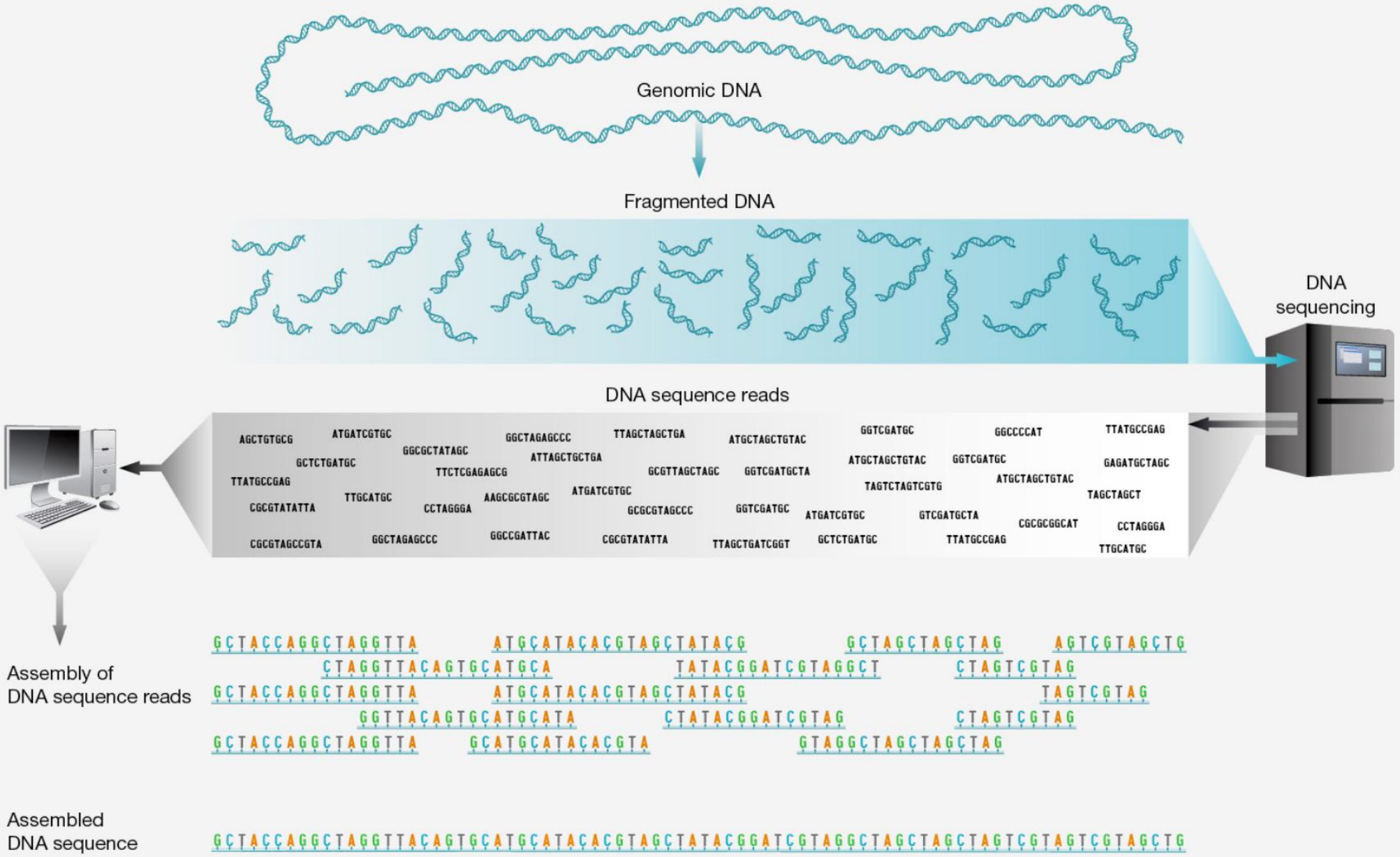


Image courtesy of National Human Genome Research Institute (https://www.genome.gov/genetics-glossary/Shotgun-Sequencing)

Bioinformatics happen here!

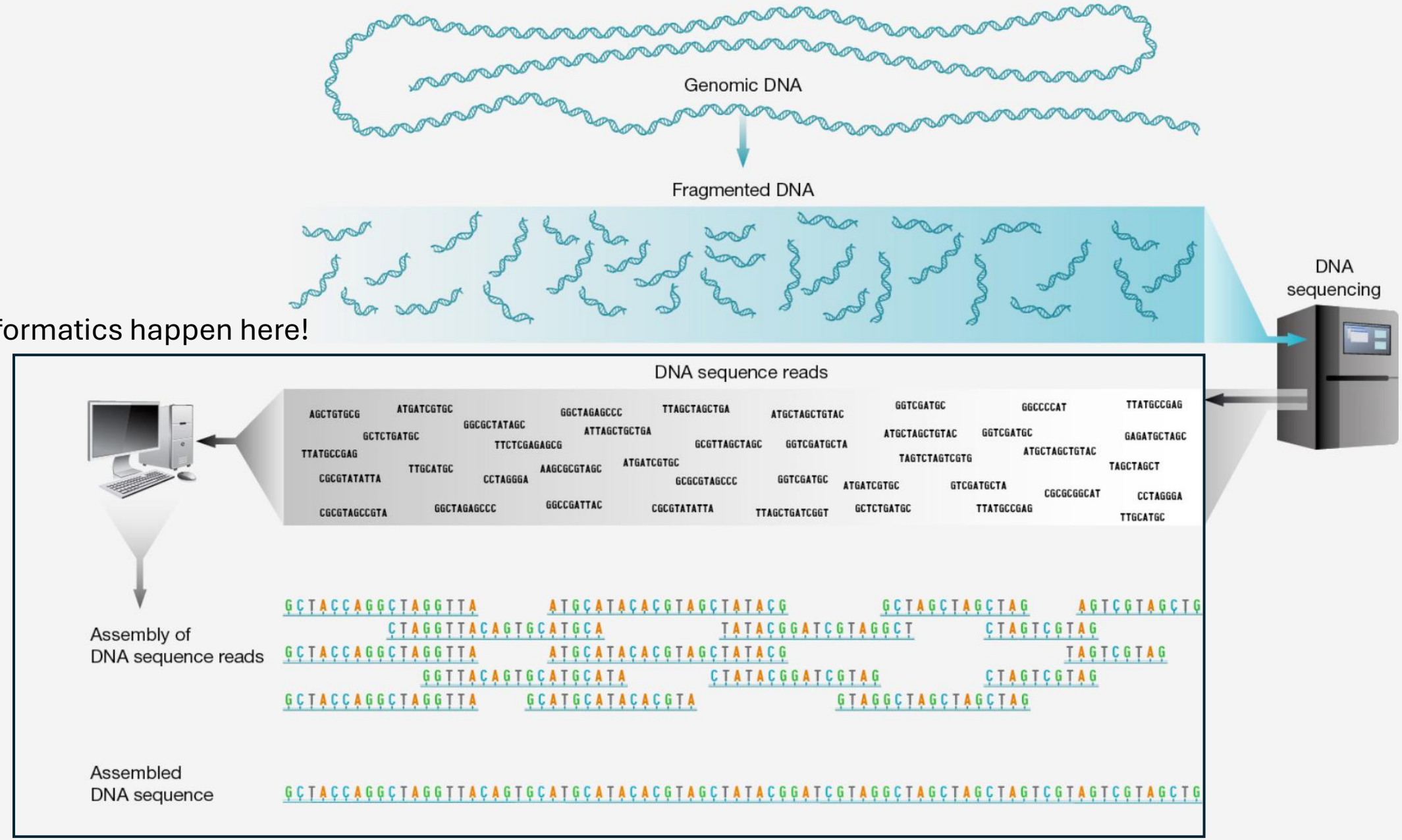
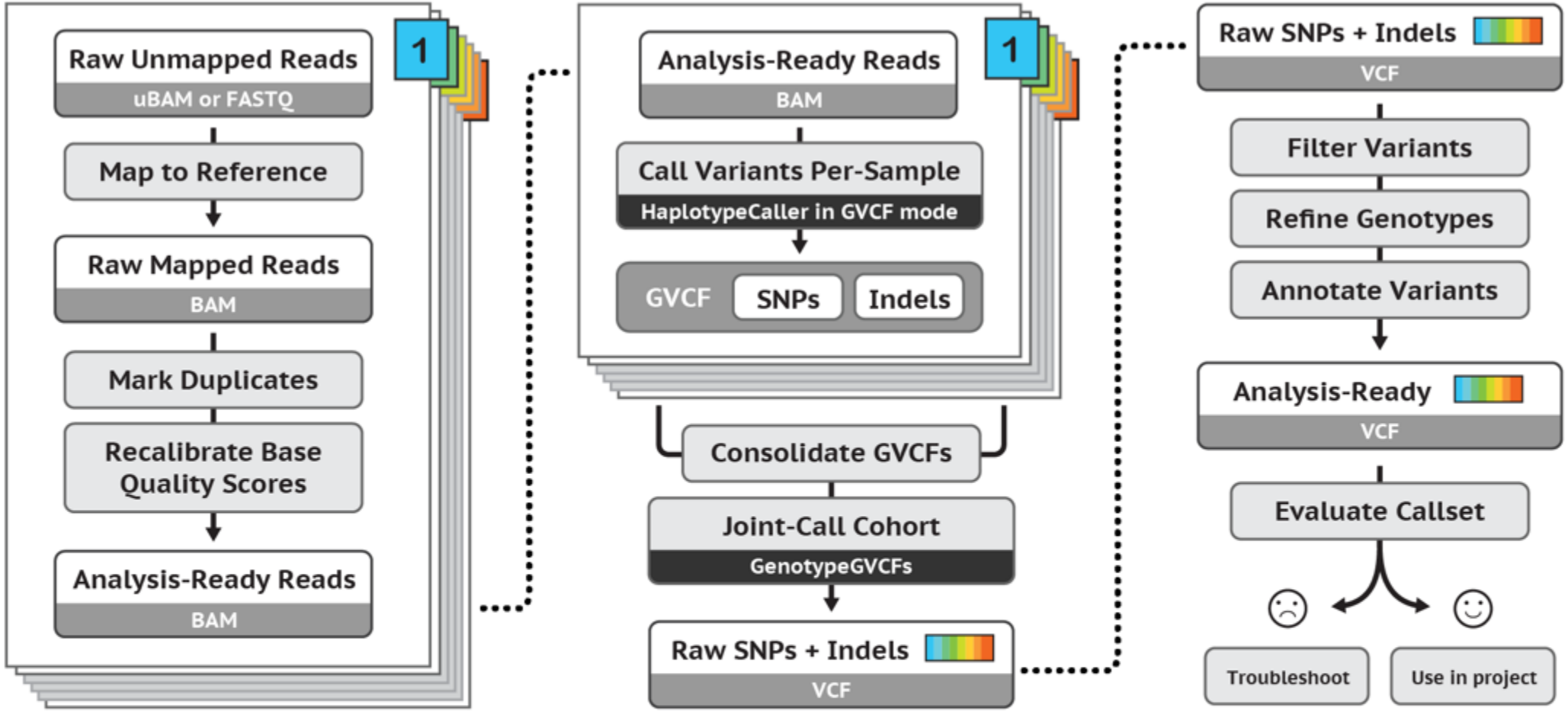


Image courtesy of National Human Genome Research Institute (https://www.genome.gov/genetics-glossary/Shotgun-Sequencing)



Best Practices for SNP and Indel discovery in germline DNA - leveraging groundbreaking methods for combined power and scalability.

https://gatk.broadinstitute.org/hc/en-us

Filtering SNPs

To reduce the number of markers, filtering is often done to remove uninformative SNPs.

- remove SNPs with poor quality
- remove SNPs with identical calls in all the samples
- based on biological functioning.

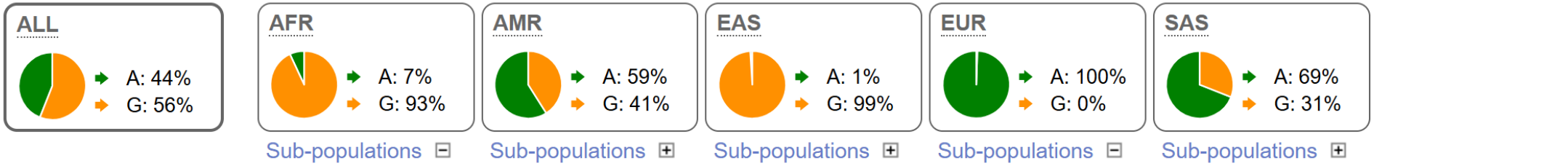
Applications of SNPs

- Genetic Ancestry and Population Genetics
- Genome-Wide Association Studies (GWAS)

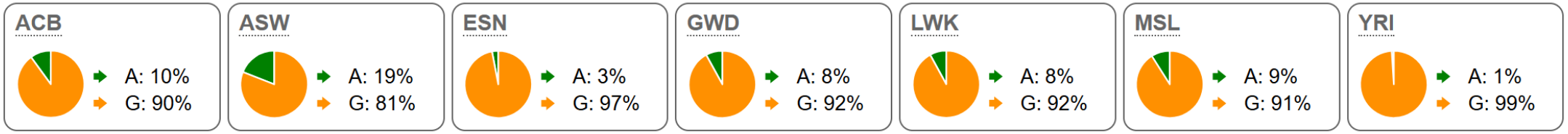
Population Structure

Population genetics

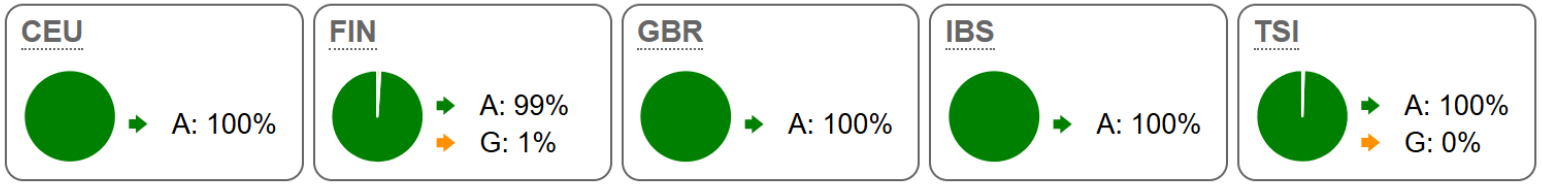
1000 Genomes Project Phase 3 allele frequencies



AFR sub-populations



EUR sub-populations

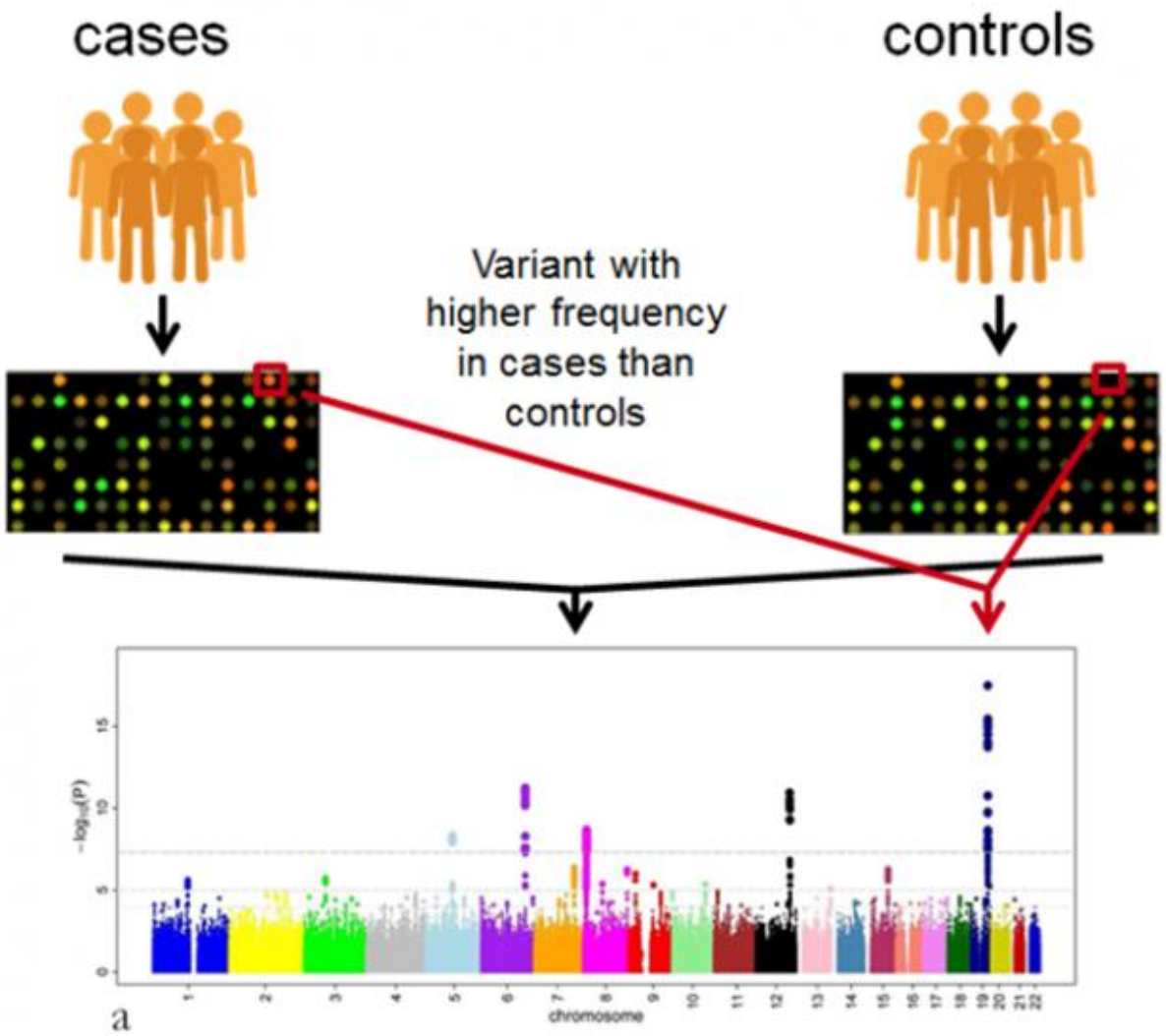


Genome-Wide Association Studies (GWAS)

Goal: Identify SNPs associated with traits or diseases

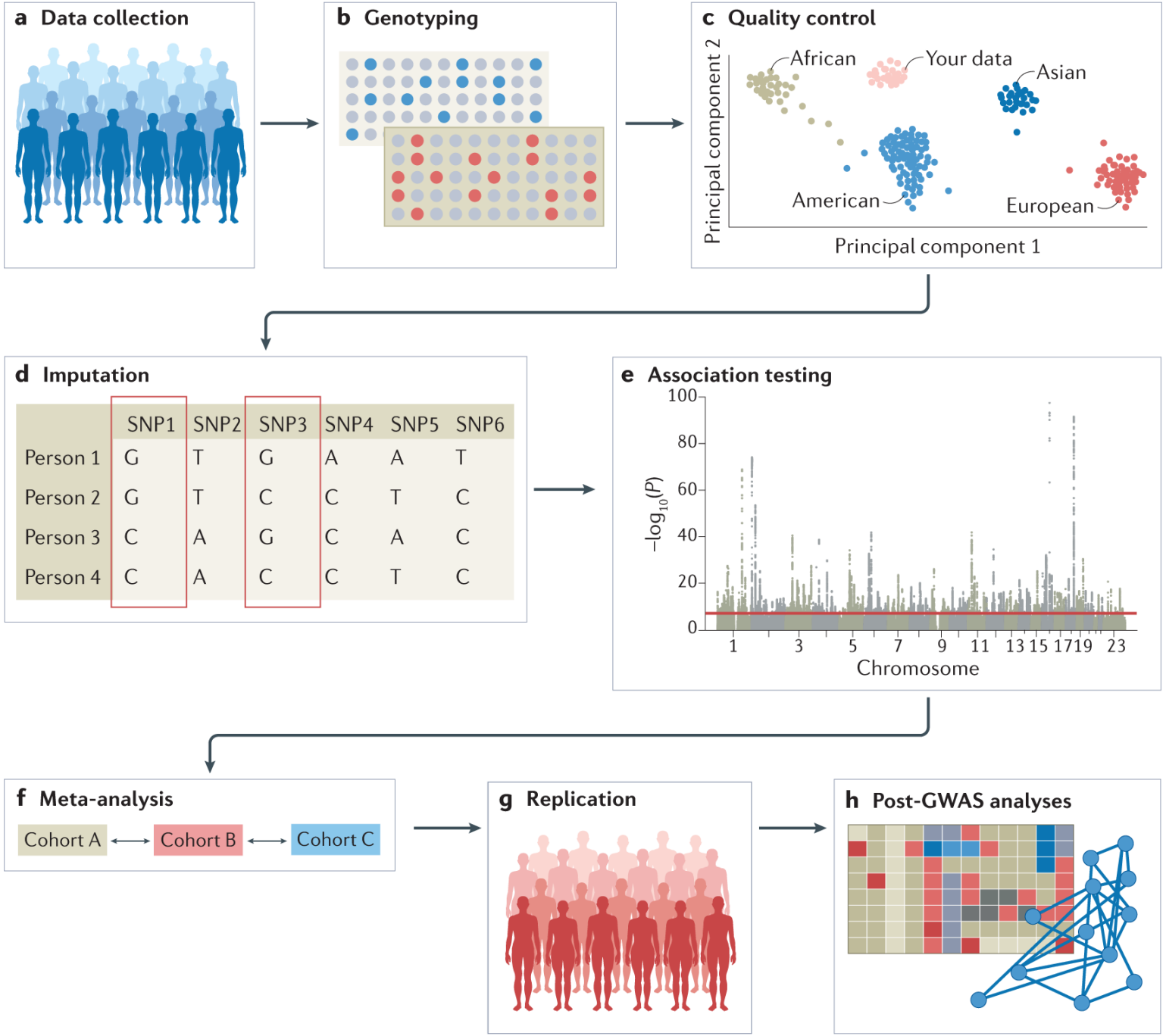
How: Compare SNP frequencies between case and control groups

Output: Manhattan plots highlighting significant associations



GWAS

Standard GWAS workflow



GWAS catalog

Exploring SNP-trait associations

The NHGRI-EBI GWAS Catalog is a publicly available resource of Genome Wide Association Studies (GWAS) and their results.

As of Feb 2025

- ***7,171*** publications
- ***786,898*** variant-trait associations
- ***106,786*** full summary statistics

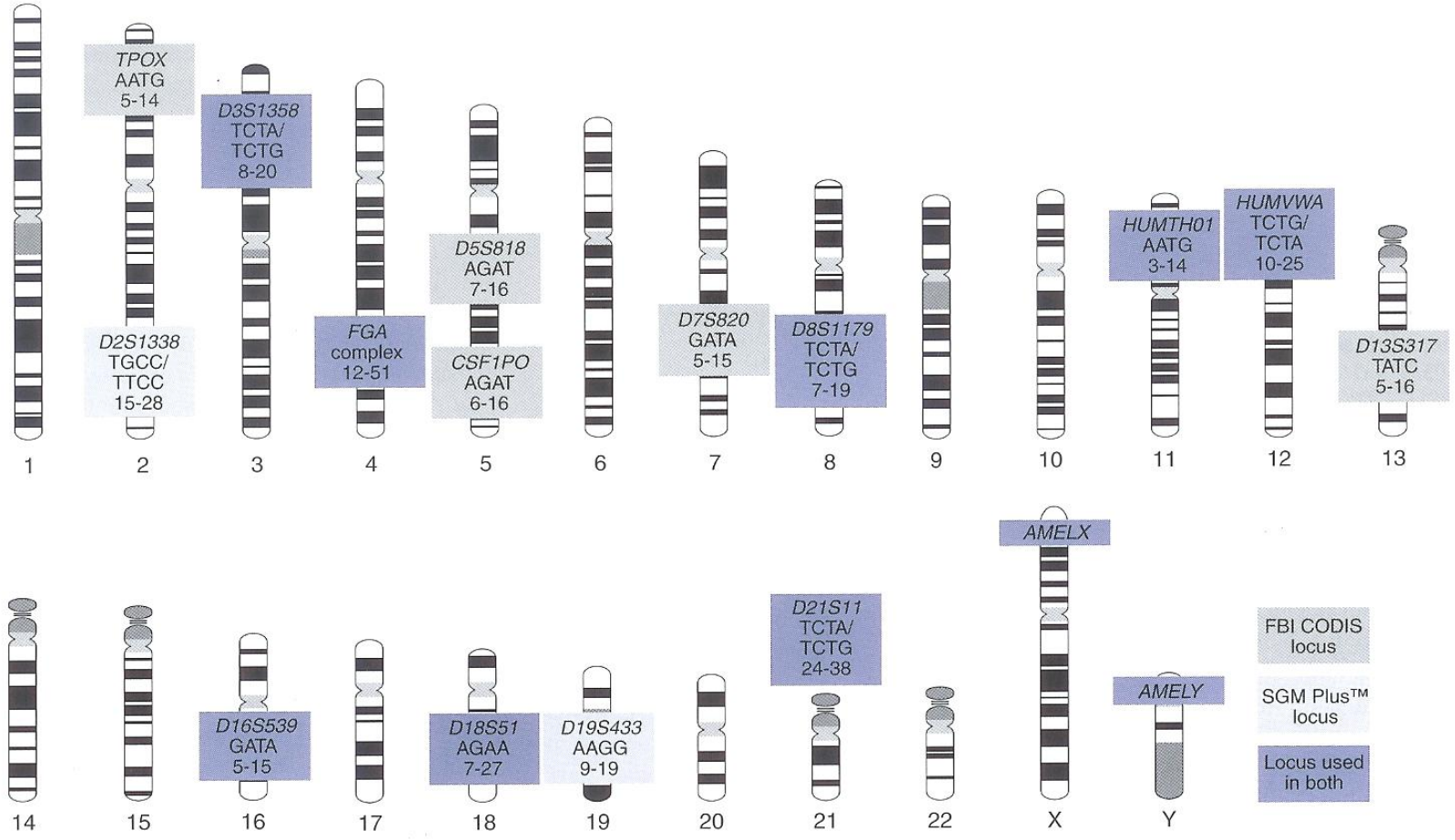
Types of Human Genetic Variants

Differences in DNA among individuals drive many types of phenotypic differences.
There are many types of genomic differences between individuals

- Structural variations (SVs)
- Copy number variations (CNVs)
- Single nucleotide polymorphisms (SNPs)
- **Short tandem repeats (STRs)**

STRs in forensics analysis

CODIS (USA) uses 20 core STR loci standardized by the FBI



https://www.mun.ca/biology/scarr/CODIS_&_SGM.html

What are STRs?

Repetitions of 1-6 bp DNA units

- Abundunt and polymorphic
- High variability in repeat number between individuals

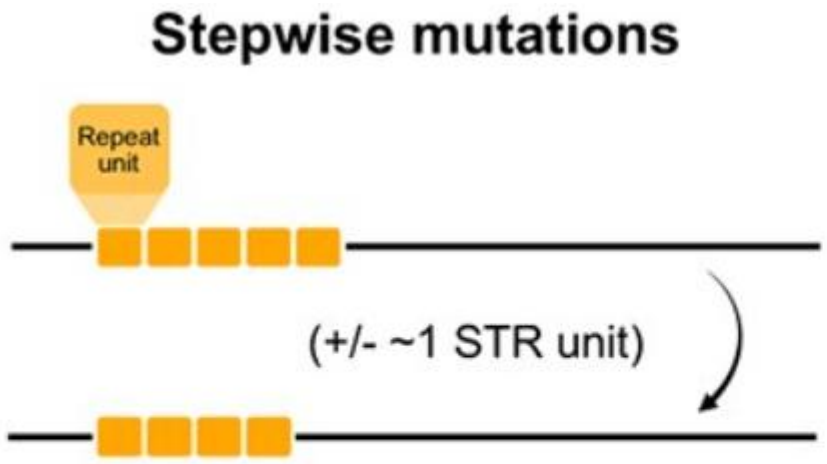
```
5' ...CGAC AT AT AT AT AT AT AT AT AT GATC...
       |||| || || || || || || || || || ||||
3' ...GCTG TA TA TA TA TA TA TA TA TA CTAG...
```

Repeat unit : AT
Repeat length : 9

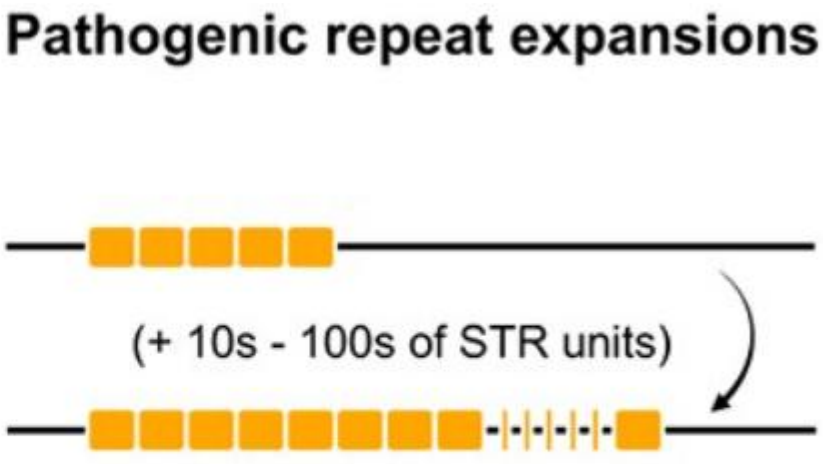
STRs vs SNPs – A Comparison

Feature	STRs (Short Tandem Repeats)	SNPs (Single Nucleotide Polymorphisms)
Definition	Repeating sequences of 1–6 base pairs	A single base change in the DNA sequence
Mutation Type	Insertion/deletion of repeat units	Substitution of one base for another
Mutation Rate	High (~10^{-3} per generation)	Low (~10^{-8} per generation)
Alleles per Locus	Multi-allelic (many allele lengths)	Typically biallelic (two possible alleles)
Size Range	Varies from 5 to 100+ base pairs	Just 1 base

STR length variations



Very common, mostly benign,
contribute to complex traits



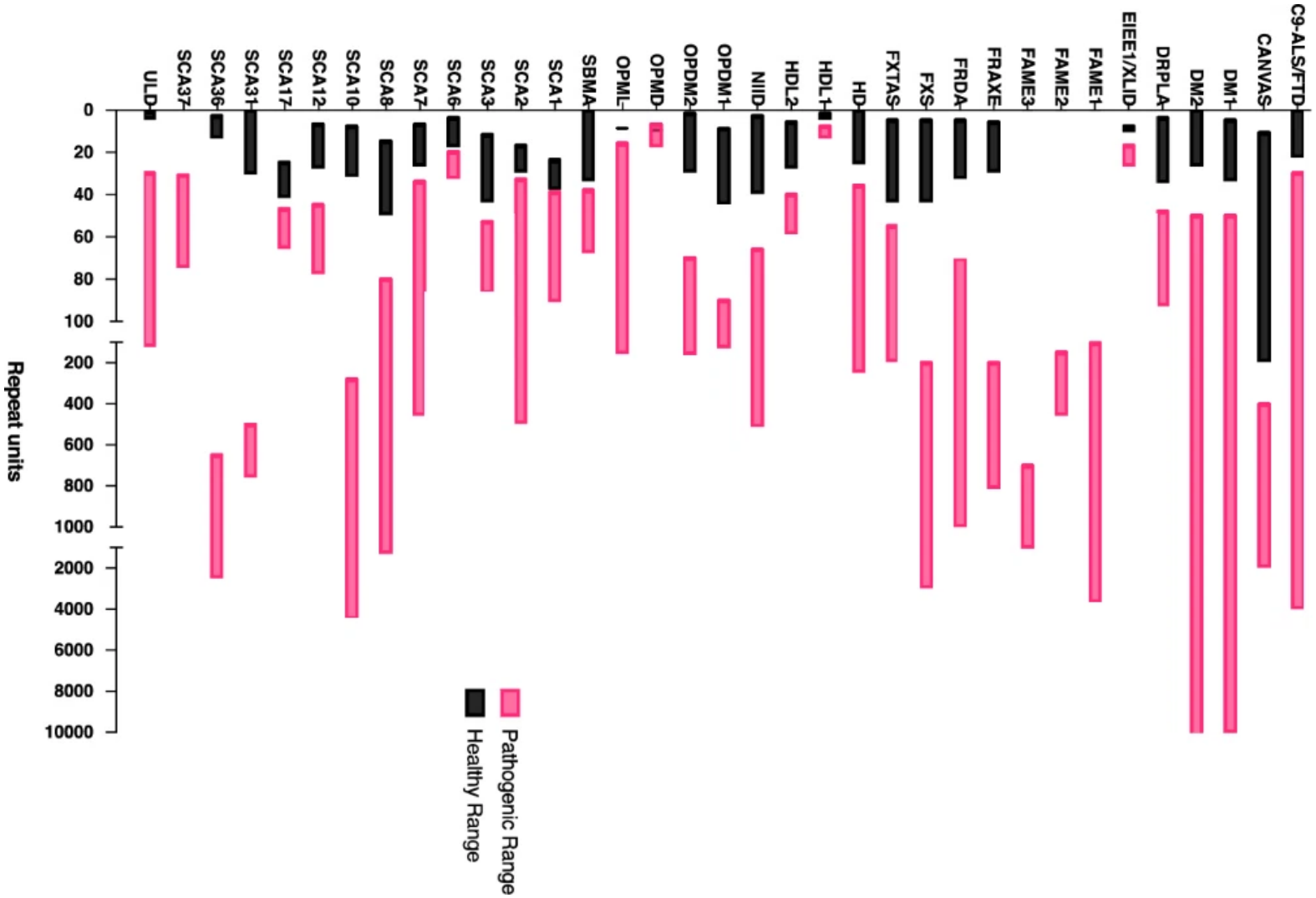
Rare, highly deleterious,
(e.g. Huntington's, Fragile X)

STRs in cancer and disease

- Microsatellite instability (MSI) is the condition of having abnormally high rates of mutation in microsatellites (STRs) within the genome, caused by a defect in the DNA mismatch repair (MMR) system. MSI tumors respond better to immunotherapy.

- Pathogenic STR expansions: Huntington's, Fragile X

Short Tandem Repeat Expansion Disorders



STR Genotyping Methods

STR genotyping refers to identifying the number of repeat units (alleles) present at specific STR loci in a DNA sample.

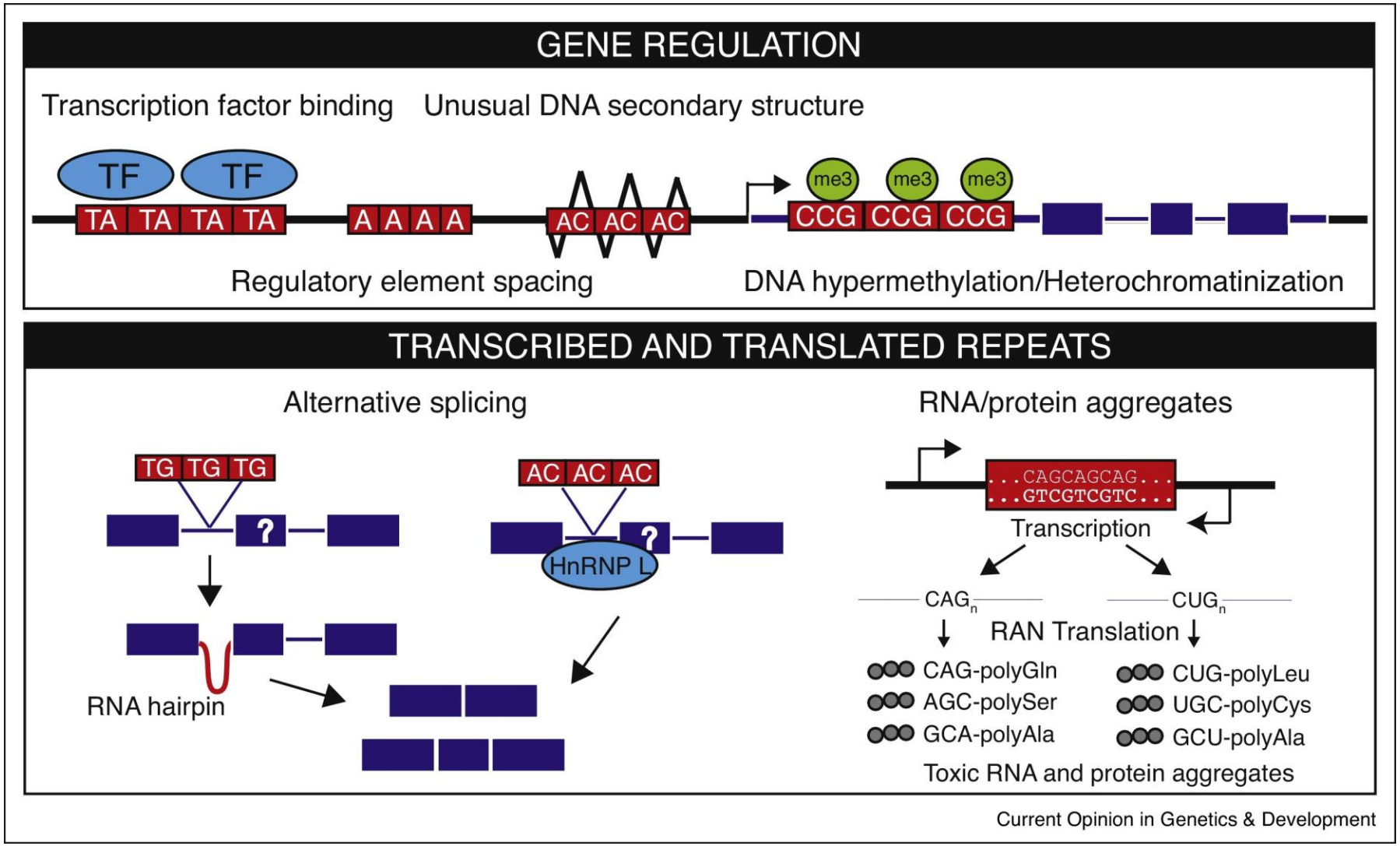
PCR

- The gold standard in forensics and paternity testing
- Microsatellite instability (MSI) testing in certain cancers

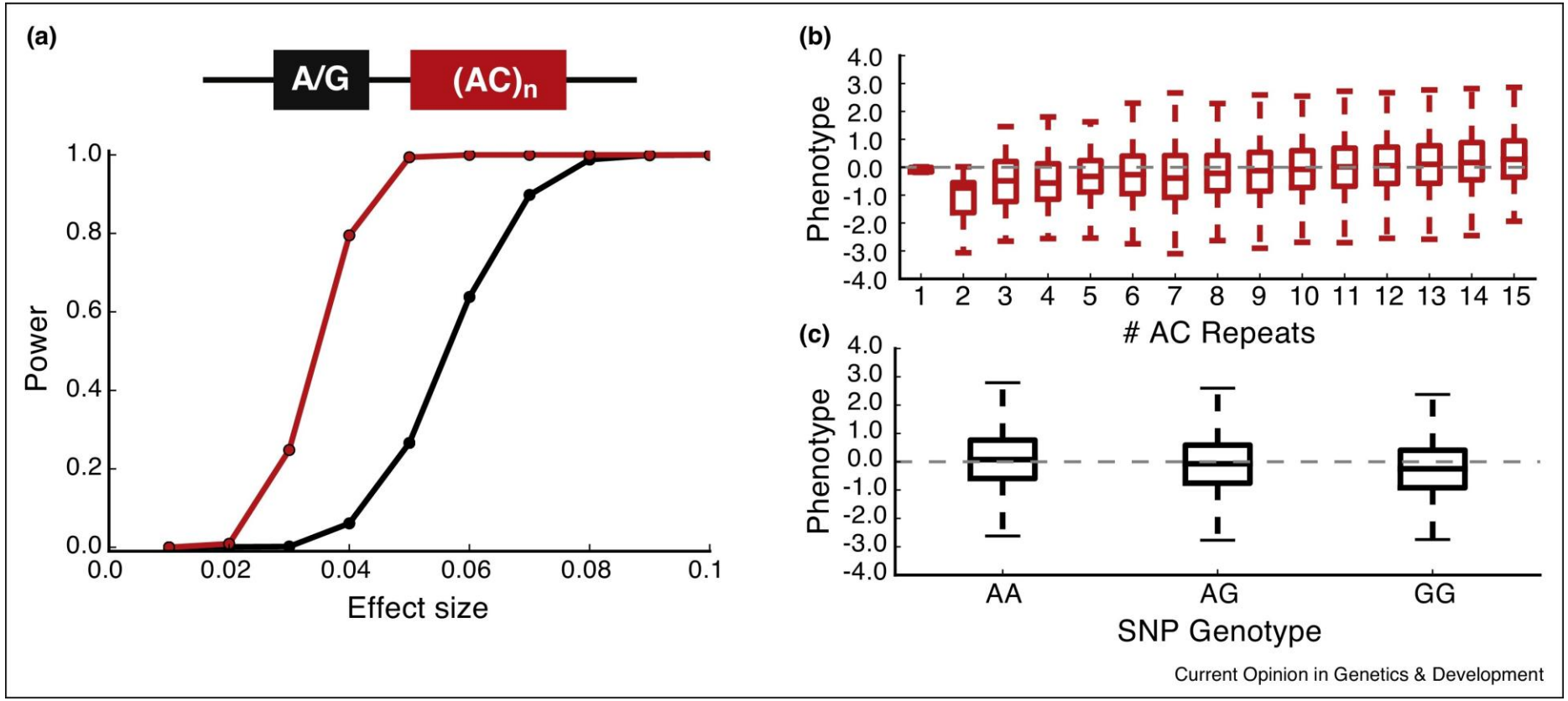
Genotyping from NGS data

- HipSTR: Handles PCR stutter and alignment errors
- ExpansionHunter: Designed for pathogenic repeat expansions (e.g., Huntington's disease)
- STRetch: Detects repeat expansions from WGS
- GangSTR: genotyping STRs longer than read length

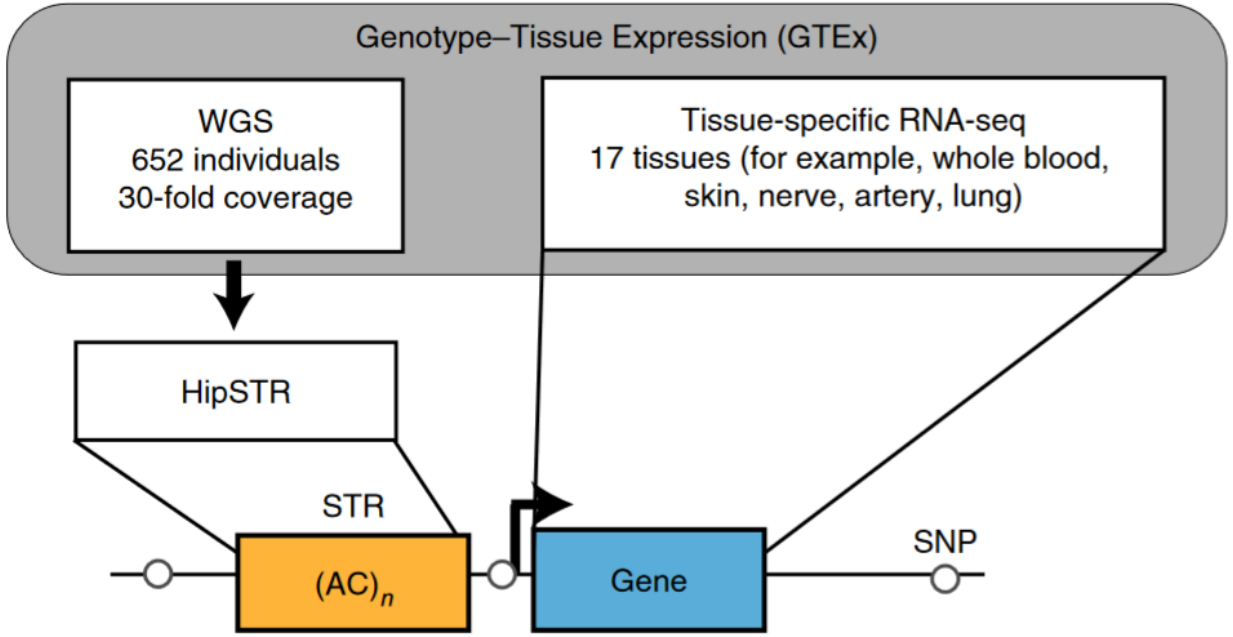
How do STRs affect gene function?



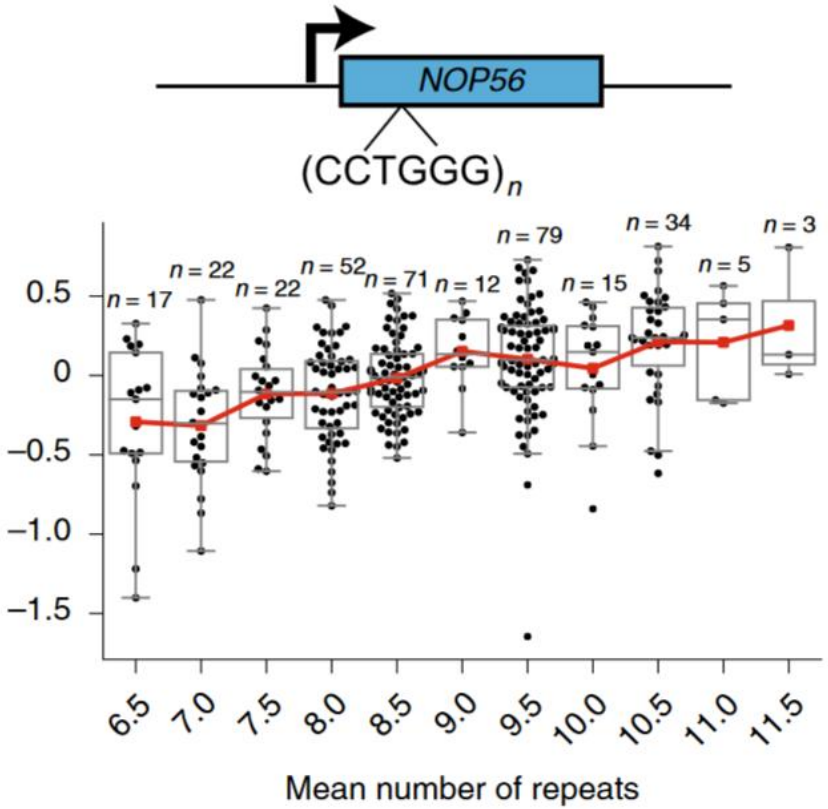
Short Tandem Repeats offer much more genotypic variation than SNPs



How do STRs affect gene expression



eSTR identification



What is population structure

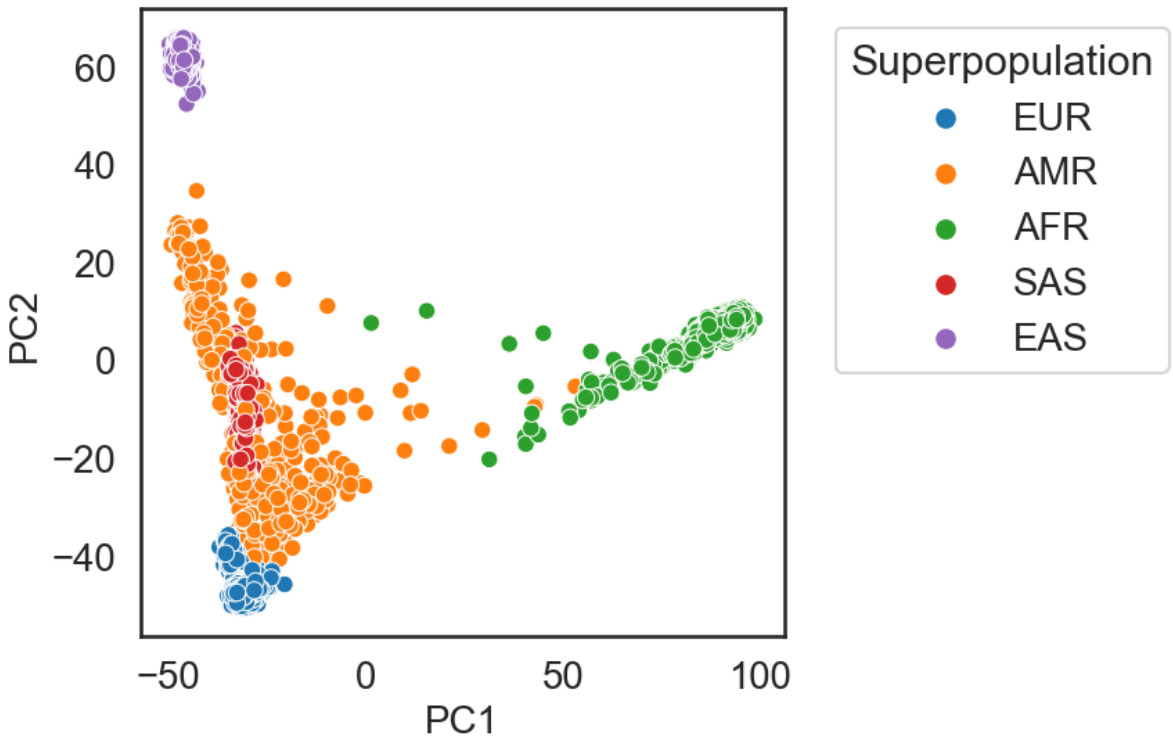
- Population genetic structure

- Individuals within a subgroup are more genetically similar to each other than to individuals in other subgroups.

- By extracting extensive variant information, such as SNPs and STRs, we can explore the genetic architecture of the population.

Methods of Population Structure Analysis

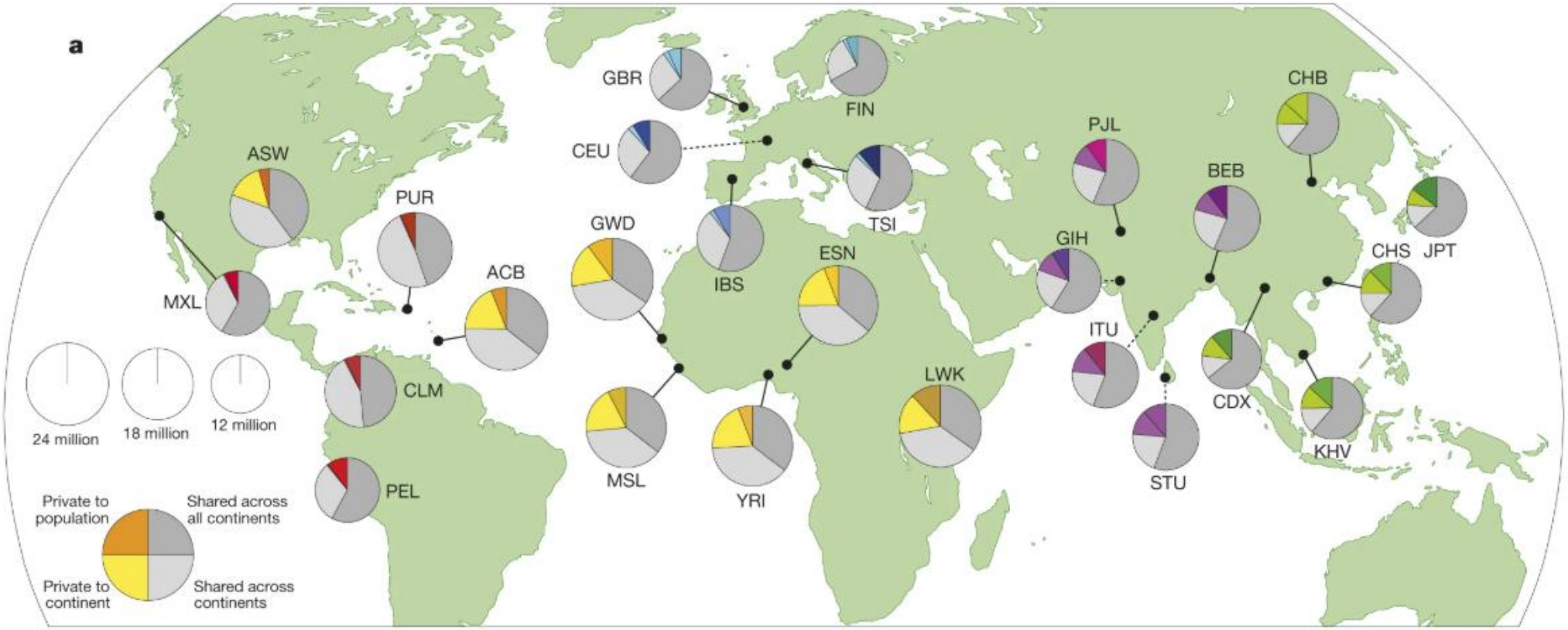
- Principal Component Analysis (PCA)



Methods of Population Structure Analysis

- ADMIXTURE

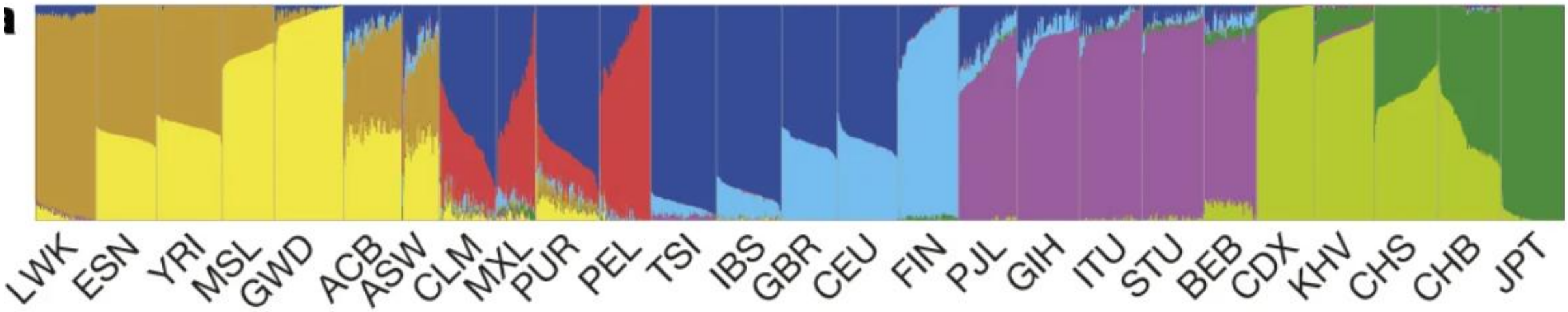
It models each individual as a mixture of K putative populations and estimate the proportions of individuals from each population.



The 1000 Genomes Project Consortium. A global reference for human genetic variation. Nature 526, 68–74 (2015). https://doi.org/10.1038/nature15393

Methods of Population Structure Analysis

Population structure inferred using the ADMIXTURE program for K = 8 from1000 genomes project



The 1000 Genomes Project Consortium. A global reference for human genetic variation. Nature 526, 68–74 (2015). https://doi.org/10.1038/nature15393

Exercise 1. Population structure analysis using SNPs

Dataset

https://ftp.1000genomes.ebi.ac.uk/vol1/ftp/data_collections/1000_genomes_project/release/20181203_biallelic_SNV/

The SNV genotypes are made available in VCF files for each chromosome. You can start with SNV on chr1.

Workflow

VCF file -- PLINK convert -- LD Pruning -- PCA analysis -- ADMIXTURE Analysis

Reading

https://link.springer.com/protocol/10.1007/978-1-0716-0199-0_4

https://connor-french.github.io/intro-pop-structure-r/

Exercise 2. Population structure analysis using STRs

Dataset

https://drive.google.com/drive/folders/1fEy09eRa0Cs4O_paZvyO5rAwnfvdt7M-?usp=sharing

The STR genotypes are available in CSV files for each chromosome. You can start with STR on one chromosome.

Workflow

CSV file -- Filtering -- PCA analysis -- Clustering analysis – Supervised classification

Reading

https://bmcbioinformatics.biomedcentral.com/articles/10.1186/s12859-024-05703-y